

APPLIED RESEARCH PROJECT · CAPSTONE PRESENTATION

A Decision-Oriented Conversational Shopping Assistant

Preference Elicitation · Explainable Recommendations · Lifecycle Support

Chenghui Tan

California State University, East Bay · April 2026

Information Retrieval $\neq$ Decision Support



Cognitive Overload

Users must mentally juggle all constraints and trade-offs themselves



Vague, Evolving Needs

Users rarely know exactly what they want when they start



Preferences Get Overwritten

Add a new constraint — your earlier choices are erased



Missing Critical Info

Delivery time, promos, and availability hidden or absent

The Decision Burden Falls on the User

*Existing tools assume shopping is a search problem.
In reality, users face a decision problem —
and no platform is built to help them solve it.*

- Decision fatigue increases with more options
- Users guess, settle, or abandon purchases
- Post-purchase regret when trade-offs aren't clear
- Trust erodes when recommendations have no rationale

→ We built a system that addresses the decision — not just the search.

Existing Platforms — Where They Fall Short

Google

Shopping

Search & filter

Amazon

Rufus

AI chat in Amazon

ChatGPT

Shopping

Natural language AI

Perplexity

AI answer engine

KEY GAPS ACROSS THE LANDSCAPE

✗ Overwrites preferences mid-conversation

✗ Assumes you already know what you want

✗ Missing delivery, promos & availability

✗ Text heavy — low information density

✗ No preference continuity across turns

✗ Multiple steps to reach product page

✗ Separates reasoning from shopping flow

✗ Opaque ranking — no explanation why

✗ Not built for first-time / cold-start users

→ The common thread: no platform treats shopping as a decision problem.

Search-first tools

→ retrieve results, not decisions

AI chatbots

→ converse, but don't structure trade-offs

All surveyed platforms

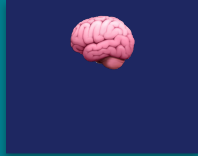
→ no preference continuity across turns

Capability Comparison Across Platforms

Capability	Google Shopping	Amazon Rufus	ChatGPT Shopping	Perplexity	Our System
Preference Elicitation	X	Partial	Partial	X	✓
Decision Modeling	X	X	X	X	✓
Explainable Recommendations	X	X	Partial	✓	✓
Delivery / Promo Info	Partial	X	X	X	✓
Preference Continuity	X	Partial	Partial	X	✓
Side-by-Side Comparison	X	X	X	X	✓
Lifecycle Support	X	X	X	X	✓

→ Our system is the only one that addresses all seven gaps simultaneously. Here is how we built it.

Your Personal AI Shopping Assistant



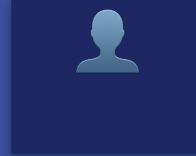
Understand You

Real Frustration → Category

Decision-Relevant Q&A

Multi-Select Use Cases

ecosystem • placement • privacy • screen size



Built for You

Preference Continuity

Visible Diff on Refine

Inferred From Free Text

“leaks” → leak-proof • “heavy” → lightweight



Clear Decisions

3 Curated Picks

Side-by-Side Compare

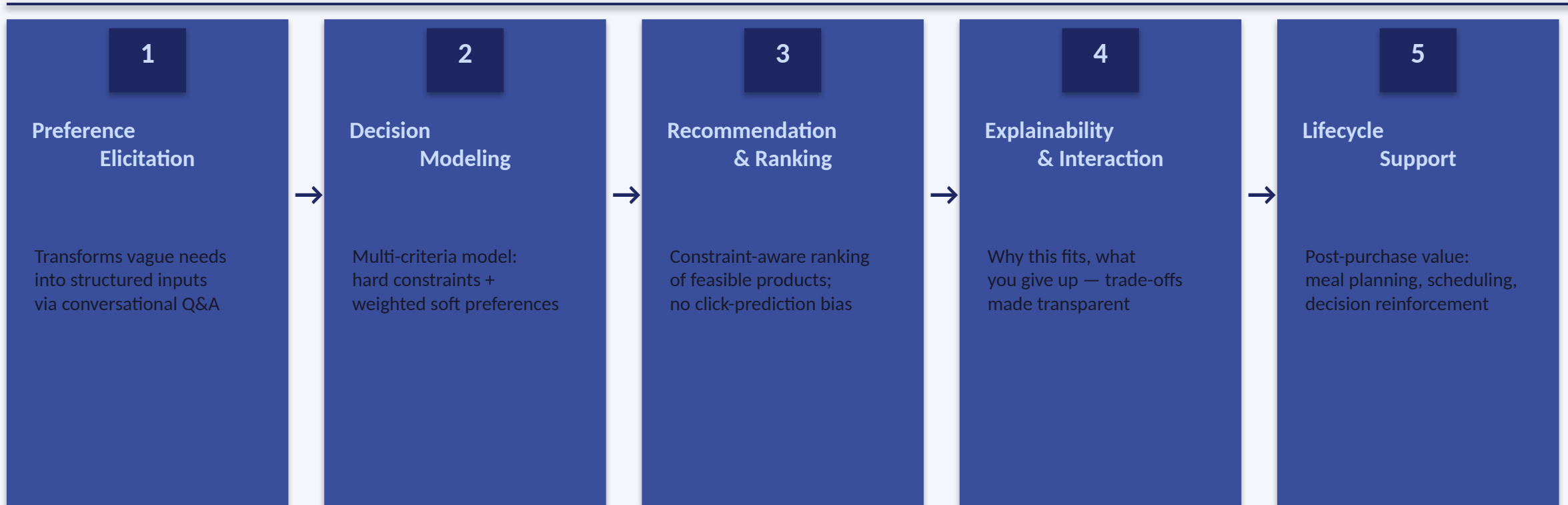
Match Checklist (✓ ✗ ?)

Best Fit • Budget Pick • Stretch Pick



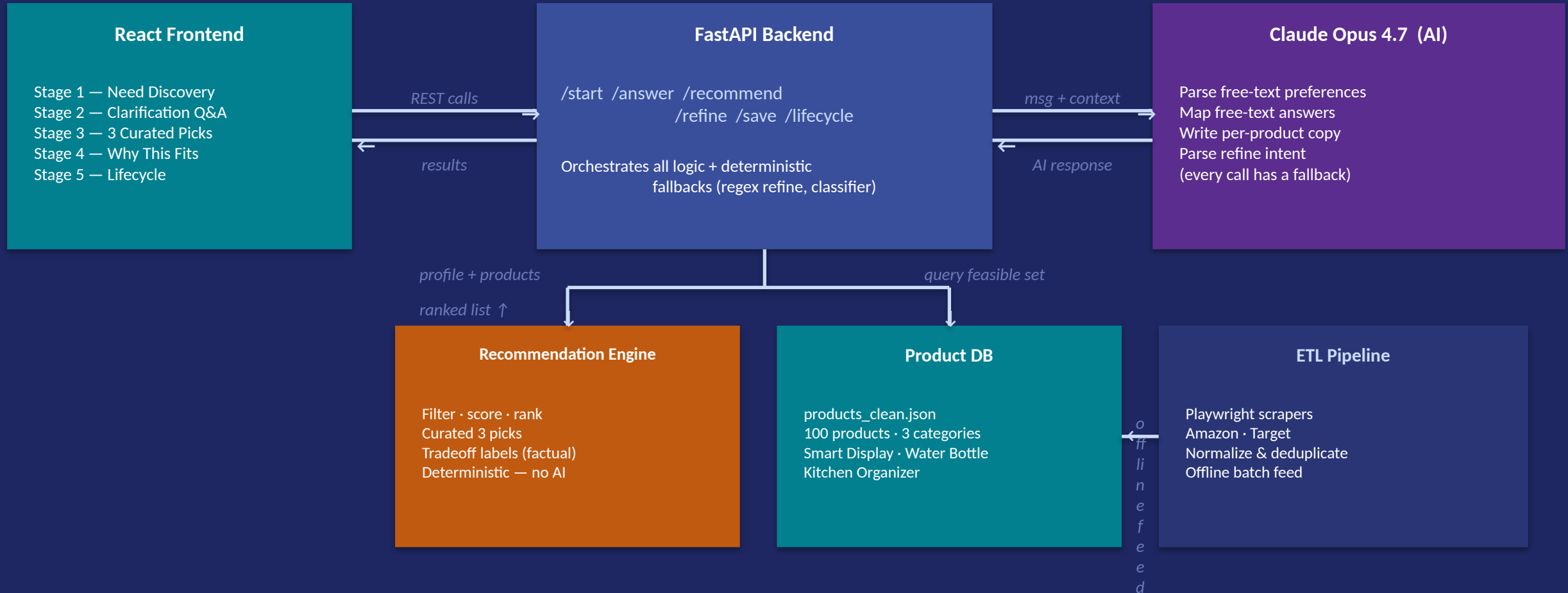
Lifecycle Support — Value continues long after the purchase

Five-Layer Decision Support Framework

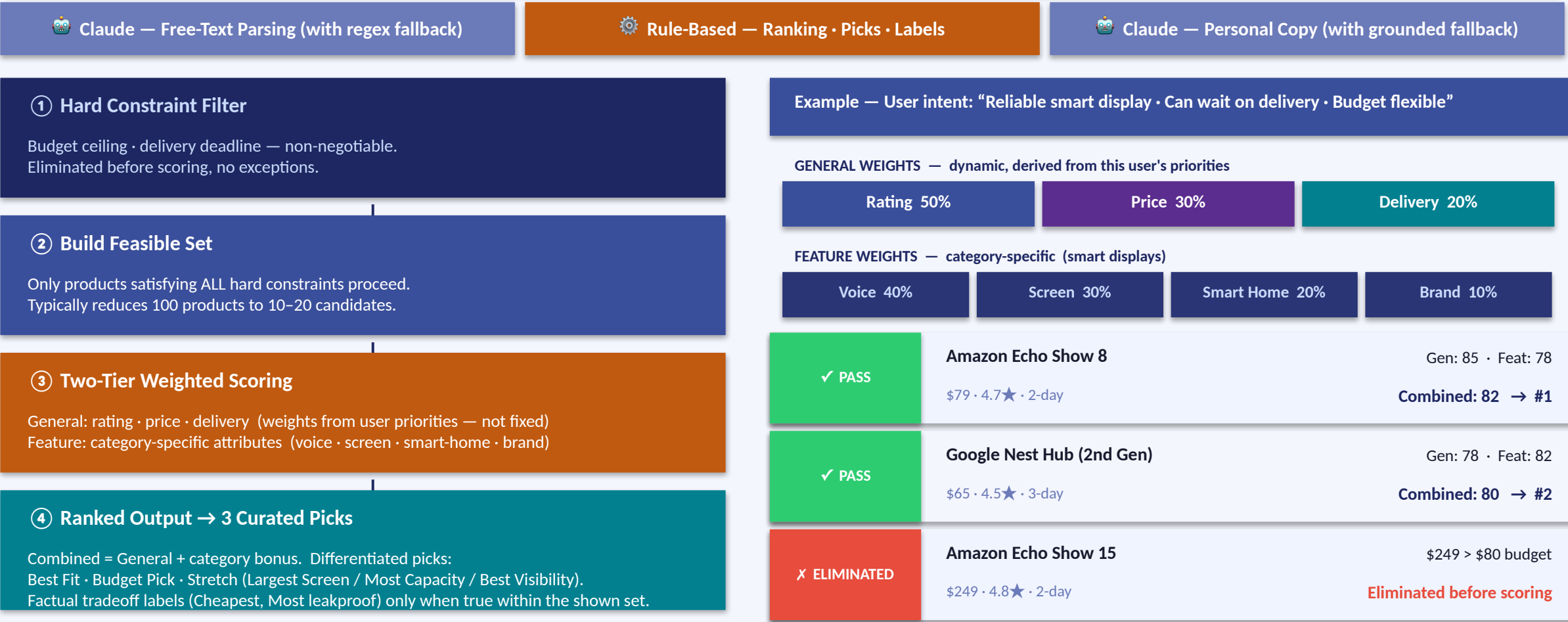


End-to-end: vague need → structured decision → ranked products → explained trade-offs → lifecycle value

Components & How They Communicate



The Deterministic Core



The Supporting Stack — What Feeds the Decision Engine

ETL + Provenance-Flagged Backfill

Playwright scrapers • Amazon • Target
Anti-bot stealth • human-like delays
Normalize: price, rating, delivery, dedup
Backfill flagged: scraped vs estimated vs synthesised

100 products • every field has provenance

Attribute Extraction (load-time)

Echo Show 5/8/15 → screen_inches • ecosystems
has_camera • mounting (wall/counter/tilt)
capacity_oz • bottle_material • drinking_style
organizer_material • organizer_visibility

feeds ranker, picks, and comparison view

Claude Opus 4.7 — 3 Call Sites

- 1 Parse free-text → category + preferences
- 2 Map free-text answers → structured values
- 3 Write personal copy + parse refine intent

Every call has a deterministic fallback —
the demo runs end-to-end without an API key.

Never on the ranking path

React Frontend — 5 Stages

- 1 Need Discovery — situational context
- 2 Clarification Q&A — 6 decision-relevant chips
- 3 3 Curated Picks — Best Fit • Budget • Stretch
- 4 Why This Fits — rule-grounded checklist
- 5 Lifecycle — realistic post-purchase support

+ Comparison overlay • Save-for-later • Diff history

Capability Validation & Scenario Walkthrough

Capability Validation

✓ Capability gaps addressed

7 of 7 gaps resolved — no competing platform addresses more than 2 simultaneously

✓ Information completeness

Decision-critical attributes surfaced inline: screen size, ecosystem, camera, capacity, material

✓ Interaction efficiency

End-to-end decision in 5 stages, zero external page navigation required

✓ Decision transparency

Every pick shows a ✓/✗/? checklist mapping each user preference to a match status — no surprises

Demo — Water-Bottle Scenario

① Need Discovery

“Current bottle is heavy and leaks in my gym bag.” → water_bottle

② Clarification (6 chips)

Gym • Stainless • FreeSip • Insulated
Large capacity • Under \$40

③ 3 Curated Picks

Best Fit Owala 24oz FreeSip • Budget Zak 19oz
Stretch Owala 32oz [Most capacity, Most leakproof]

④ Why + Checklist

5 rule-grounded reasons + ✓/✗/? per chip.
Budget Pick honestly shows ✗ Large capacity (19oz).

⑤ Refine + Compare

‘larger’ → flips Best Fit to 32oz.
Compare 3 finalists side-by-side.

Honest Assessment • What This Proves • Next Steps

Limitations

- ⚠ Small dataset — 100 products, 3 categories only
- ⚠ No formal user study — heuristic evaluation only
- ⚠ 72/100 delivery values are estimated (flagged in data)
- ⚠ Rule-based ranking — weights inferred, not learned

Conclusions & Impact

- ✓ Decision support feasible end-to-end without an LLM key
- ✓ Explanations grounded in the same rules the ranker used
- ✓ Visible preference continuity — every refine shows a diff
- ✓ Honest about what's possible — realistic post-purchase copy
- ✓ Generalisable: same pattern across all 3 categories

Future Work

- Formal user study (SUS, task completion time)
- Live retailer API for real-time pricing
- ML-based ranking from interaction data
- Expand to more product categories
- Mobile-first UI redesign

Thank You

Questions & Discussion

Author:	Chenghui Tan
Institution:	California State University, East Bay (CSUEB)
Project:	Decision-Oriented Conversational Shopping Assistant
Stack:	React · FastAPI · Claude Opus 4.7 · Python · Playwright (ETL)